MILITARY SPECIFICATION

FILTER-SEPARATOR, LIQUID FUEL FFU-25/E

1. SCOPE

1.1 This specification covers a 1,200-gpm-capacity aircraft liquid fuel filter-separator, designated FFU-25/E.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-104	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods of
MIL-F-8901	Filter-Separators, Aviation and Motor Fuel, Ground and Shipboard Use, General Requirements and Test Procedures for
MIL-V-11201	Valve, Safety Relief, Angle, Brass or Bronze, Extra-Heavy, For Gasoline Service
MIL-F-52308	Filter Element, Fluid Pressure

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-808	Finishes, Protective, and Codes for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-831	Test Reports, Preparation of

FSC 4330

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society of Mechanical Engineers

Boiler and Pressure Vessel Code

Section VIII Pressure Vessels
Section IX Welding Qualifications

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, New York 10017.)

American National Standards Institute, Inc

B16.5 Steel Pipe Flanges and Flanged Fittings

(Application for copies should be addressed to the American National Standards Institute, Incorporated; 10 East 40th Street; New York, New York 10016.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The filter-separator shall consist of the following major components:

Item Name	See Requirement
Housing	3.7
Canisters and elements	3.8
Water slug and flow control valve	3.9

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3.3 General specification. The requirements of MIL-F-8901 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.4 Materials. Materials shall be as specified herein and in MIL-F-8901.

3.5 Design and construction. The filter-separator shall meet the design requirements of MIL-F-8901 at a flow rate of 1,200 gpm.

3.6 Performance. The filter-separator (assembled with elements and all components) shall be capable of meeting the performance requirements of MIL-F-8901 at a flow rate of 1,200 gpm.

3.7 Housing. The configuration of the filter-separator housing shall be as shown on figure 1. The housing shall be fabricated from aluminum-alloy material of the grade and temper to suit the application. The head shall be sealed to the body by means of an O-ring mounted in a circular groove at the point of closure. The unit shall be constructed and labeled in accordance with sections VIII and IX of the ASME Boiler and Pressure Vessel Code.

3.7.1 Working pressure. The housing shall be designed for a working pressure of 150 psi.

3.7.2 Cartridge stools. Cartridge stools shall be provided to accommodate the filter elements. The stools shall be smooth and shall have an outside diameter of 1.184 to $\pm$0.002 inch.

3.7.3 Lifting devices

3.7.3.1 Head lifting device. The housing shall be of the vertical type and shall be equipped with a hinged or pivoted device to facilitate lifting the head and swinging or rotating it to one side to permit servicing the filter-separator. The hinge or pivot shall support the head during servicing without distortion or misalignment of the head or components. Unless otherwise specified (see 6.2), swing-type bolts shall be used on all main closures.

3.7.3.2 Lifting eyes. The housing shall be provided with 3-inch-diameter lifting eyes suitably spaced for stability during lifting and designed to support a weight of 2-1/2 times the gross weight of the filter-separator.

3.7.4 Inlet and outlet connections. Inlet and outlet connections shall be 8-inch nominal pipe size and shall be located in line and at the same elevation essentially as shown on figure 1. Both connections shall be provided with flange connections, faced and drilled in compliance with ANSI Standard B16.5, class 150.

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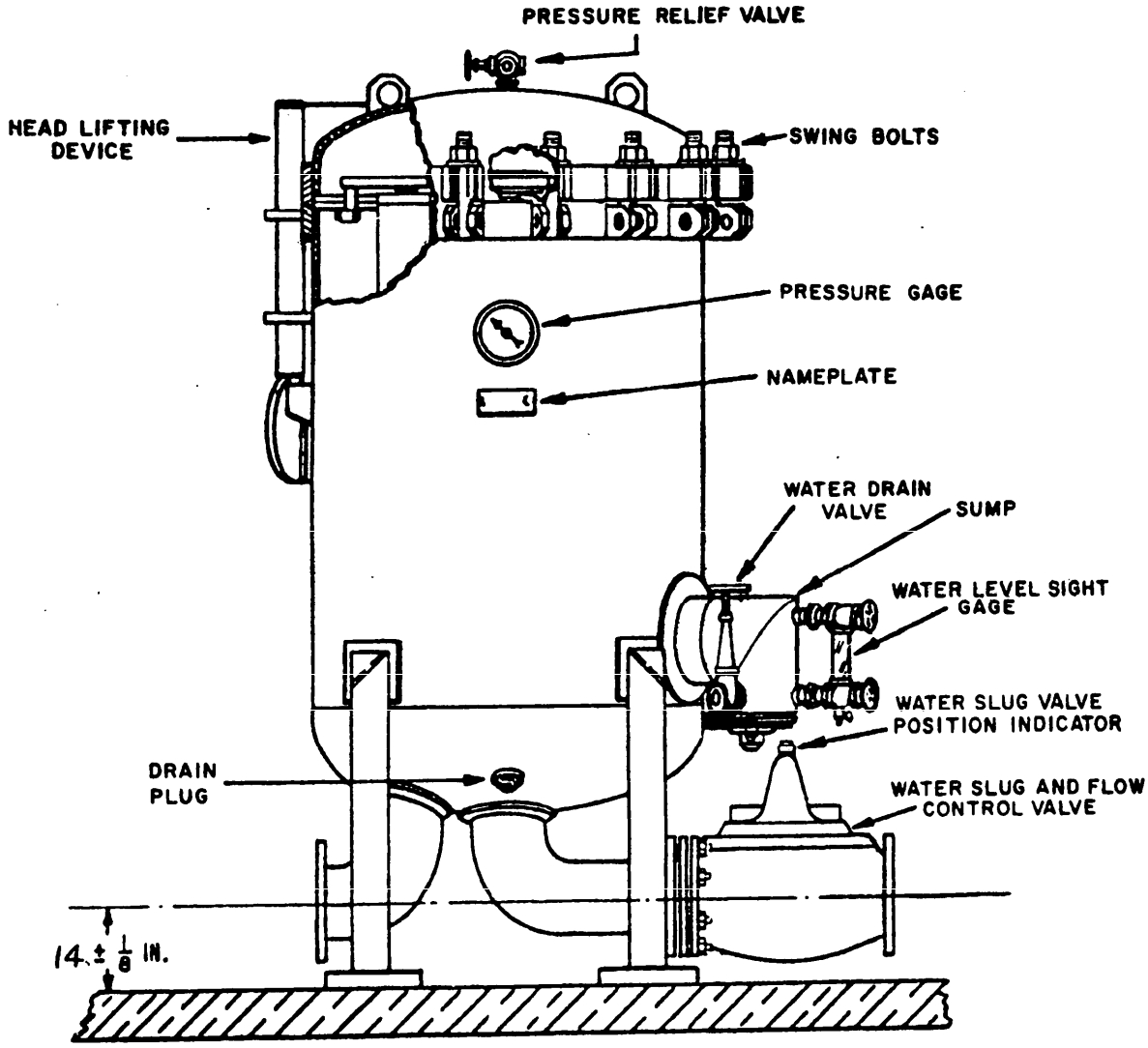


FIGURE 1. FILTER-SEPARATOR

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3.7.5 Automatic water drain valve. An automatic water drain valve shall be incorporated into the housing to automatically permit the removal of all water arrested by the filter-separator. The draining capacity shall be in excess of 60 gpm at internal pressures from 30 to 150 psi. The valve shall be either of the single- or double-chamber type. The drain shall be so designed that should the diaphragm fail, the valve disc will remain in the closed position. The valve shall be installed in a separate sump on the side of the housing.

3.7.5.1 Float control tester. A float control tester shall be incorporated into the valve design. The tester shall provide a means to manually actuate the float control pilot to periodically check the operation of the fuel discharge and water drain valve.

3.7.6 Air eliminator and pressure relief valves. A class 2, size 1-1/4-inch angle pattern pressure relief valve conforming to MIL-V-11201 shall be provided on top of the vessel as shown on figure 1. An automatic air eliminator shall be incorporated into the design of the filter-separator. It shall be so designed that all air will be released from the filter-separator during startup with a minimum of fuel leakage. The pressure relief valve shall be set to open at 165 psi.

3.7.7 Differential pressure gage (bellows type). A bellows-type differential pressure gage shall be supplied. The gage shall be equipped with liquid-filled, ruptureproof bellows and a torque tube. The case shall be at least 6 inches in diameter and shall be of equal construction to that of the die cast aluminum, finished with a weather-resistant black epoxy resin enamel. The case shall be provided with a glass cover, secured to the bezel with an O-ring seal. The face of the gage shall be provided with white numbers on a black background, or black on white. The housing shall be designed for a system working pressure of 150 psi. The indicator shall have a uniform range of from 0 to 30 psi with a scale graduation of 1-psi increments. The gage shall be designed for panel mounting with 1/4-18 NPT internal thread connections. The inlet port shall be clearly marked INLET, and the outlet port clearly marked OUTLET. The accuracy shall be $\pm$1/2 percent of the full-scale differential pressure.

3.7.7.1 Gage panel. The differential pressure gage shall be attached to the filter-separator by a gage panel.

3.7.7.2 Differential pressure gage valve manifold. To facilitate removal of the differential pressure gage for periodic calibration, a Barton Corporation model 235 valve manifold, or equal, shall be provided. This manifold shall be connected to the differential pressure gage and shall be installed on the gage panel in a manner to permit opening and closing of all valves. The manifold shall be made from brass and shall incorporate two block valves and one bypass valve. The gage connections shall be 1/4 inch female NPT. The

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manifold shall be capable of withstanding static pressures up to 500 psi without failure. All valves shall be packless type and shall incorporate stainless-steel diaphragms. Preformed tubing and tubing fittings shall be provided to facilitate assembly.

3.7.8 Drain valve. The filter-separator shall be equipped with a 1-1/2-inch manually operated water and fuel drain valve. The valve shall be a bronze or aluminum-alloy ball-type valve suitable for the application. The valve shall drain all water, fuel, and sediment from the filter-separator by gravity. The valve shall be installed on the water sump as shown on figure 1.

3.7.9 Water level sight gage. A 1/2-inch, armoured unbreakable, clear plastic liquid level gage shall be provided for observing the water accumulation at the bottom of the sump. The gage shall be equipped with nickel-copper-alloy ball checks in both the upper and lower fittings, an upper and lower shutoff valve, and a bottom blowoff cock.

3.7.10 Sampling device. A quick-release valve and plug assembly (Millipore Filter Corporation catalog part No. XX 64 037 03, or equal) shall be installed in the outlet side of the filter-separator. The samples shall be capable of accepting a turbofuel sampling kit for drawing the samples required to assure fuel quality.

3.7.11 Fuel drain plug. The housing shall be provided with a 1-1/2-inch NPT fuel drain plug.

3.7.12 Legs. Four angle-shaped legs shall be welded to the housing. Each leg shall be fitted with a base plate.

3.7.13 Element access foot ledge. Approximately 4 feet from the top of the housing, two element access foot ledges shall be welded to the outside of the housing. The ledges shall be provided to permit personnel to stand on either side of the housing to change elements. Each ledge shall be 1 foot wide, 2 feet long, and 3/8-inch thick.

3.8 Canisters and elements. The filter-separator shall be provided with 30 canisters. Each canister shall contain two elements conforming to MIL-F-52308. The canister shall be as shown on figure 2.

3.8.1 Holding device. The filter-separator shall contain a spider assembly or other means of holding the canisters in position. The spider or holding device shall be designed for adjustment, removal, and replacement of canisters and elements.

3.8.2 Screen. The canister shall be provided with a tetrafluoroethylene-coated, 100- or 200-mesh steel screen to repel and separate water from the fuel.

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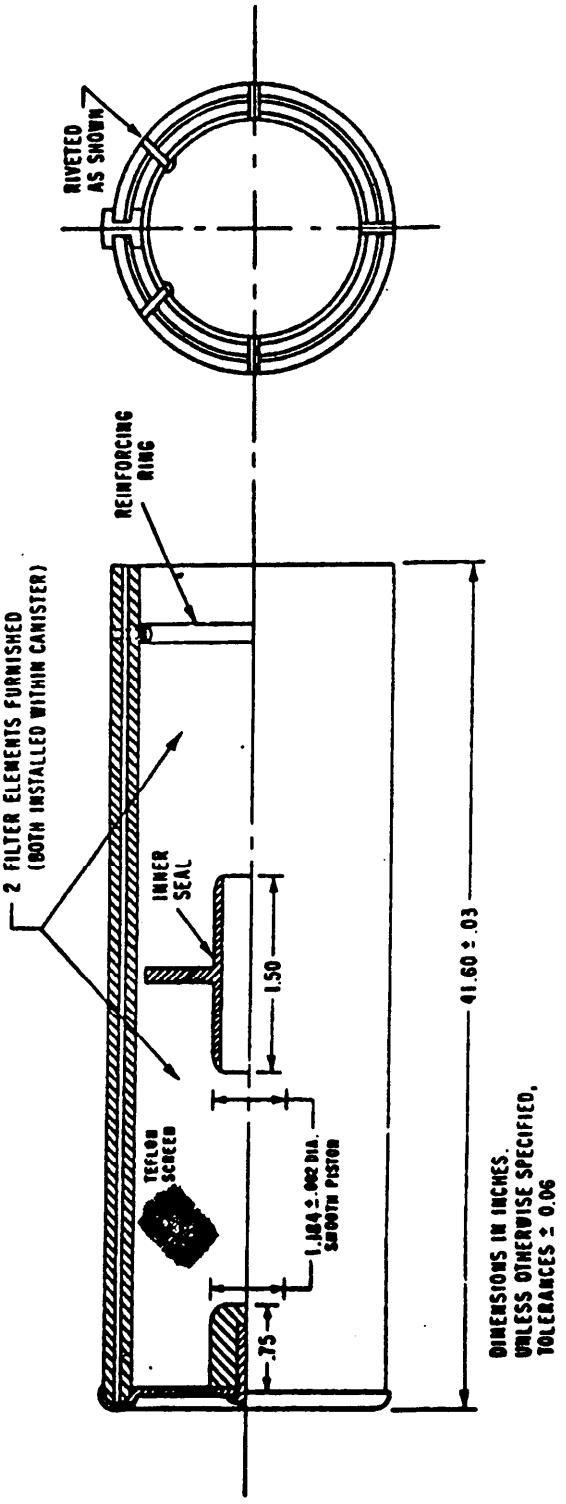


FIGURE 2. Canister Assembly

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3.9 Water slug and flow control valve. The filter-separator shall be fitted with a water slug and flow control valve installed on its outlet. The body of the valve shall be fabricated from 356-T-6 aluminum with stainless-steel trim and suitable for a maximum surge pressure of 275 psi. This valve shall limit the flow through the filter-separator to 1,200 gpm. The valve shall automatically stop fuel discharge from the unit when the incoming fuel is contaminated with a quantity of water exceeding the draining capacity of the automatic drain valve. It shall be designed to permit normal operation of the filter-separator automatically, after the water draining period, which follows closure of the valve due to a water slug.

3.9.1 Water slug valve position indicator. The water slug and flow control valve shall be equipped with a position indicator to provide visual means to determine whether the slug valve is open or closed.

3.10 Finishes and protective coatings. Cleaning, painting, plating, and chemical films shall be in accordance with MIL-STD-808. The finish shall be in accordance with finish D-351 of MIL-STD-808.

3.11 Marking of inlet connection. The word INLET shall be plainly stamped or embossed in metal adjacent to the inlet connection.

3.12 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The main equipment nameplate shall be mounted on the housing, and in addition to the usual MIL-STD-130 requirements, shall include the following markings in letters 3/32-inch high or larger:

Filter-Separator, Liquid Fuel FFU-25/E
Design Flow Rate - 1,200 GPM
Operating Pressure - 150 PSI
Elements
  First Stage
    QTY - 60
    FSN 4330-983-0993
    OD 3-3/4 inches
  Second Stage
    Permanent teflon screen
  Contract No.*
  Manufacturer*
  Specification MIL-F-83402 (USAF)
  Change elements at a differential pressure of 15 psi,
    or 3 years, whichever occurs first.
  Date elements installed**

*Applicable information shall be entered by the contractor.

**Applicable raised blocks without finish shall be provided for information to be written in by Government personnel.

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3.13 Workmanship. The filter-separator, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws and bolts; cleaning, et cetera.

3.13.1 Cleaning. Components of the filter-separator shall be thoroughly cleaned to remove dirt; excess soldering, brazing, and welding flux; welding slag; loose, spattered, or excess solder; metal chips; and other foreign materials before, during, and after assembly.

3.13.2 Riveting. Riveting operations shall be performed to insure tight and satisfactory heading of rivets.

3.13.3 Gear and lever systems. Gear and lever systems shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, and other irregularities. Where required for accurate adjustments, gear assemblies shall be free of visible backlash.

3.13.4 Clearances and adjustments. All parts such as shafts, pinions, pistons, gears, valves, et cetera, shall have proper clearances and adjustments. They shall be designed and constructed to withstand the conditions met in shipping, storage, installation, and service.

3.13.5 Bolted connections. Bolts and nuts shall be fabricated from 2024-T4 milled aluminum with alumilite finish. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.13.6 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall develop strength and transmit stress to the parts connected under proof and service loading without failure.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection and testing. The inspection and testing of the filter-separator shall be conducted on a full-scale test system in accordance with MIL-F-8901 and as specified herein.

4.2 Preproducting testing

4.2.1 Test sample. Unless otherwise specified (see 6.2), the preproduction test sample shall consist of a complete filter-separator, 183 additional

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elements, and one additional new canister all representative of a production lot. The filter-separator, canister, and elements shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.2.1.1 Data to accompany test samples. The preproduction test samples shall be accompanied by the following data:

a. Brief operating and installation data to enable test personnel to correctly install and operate the equipment

b. Certification that the filter-separator passed the Hydrostatic pressure tests as specified in 4.4.3 and MIL-F-8901

c. Engineering data in the form of assembly drawings (two sets).

4.2.1.2 Test sample and data for the procuring activity. When tests are conducted at a site other than the laboratory of the procuring activity, the following shall be available to that activity:

a. Three copies of a test report in accordance with MIL-STD-831. In addition to results, the report shall contain complete records of the tests, including data sheets, performance curves, chronological test records, photographs, sample calculations, test procedures, and a description of the test apparatus.

b. A sample element and canister removed from the filter-separator which was subjected to the preproduction test

c. A color photograph of the sample element

d. A new filter element

e. A new canister.

4.2.2 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.4. If elements supplied are qualified under MIL-F-52308, preproduction tests may be waived.

4.3 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests . . . . . See 4.3.1

b. Sampling plan and tests . . . . . See 4.3.2.

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4.3.1 Individual tests. Each filter-separator shall be subjected to the following tests:

a. Examination of product . . . . . See 4.4.2

b. Canister inspection . . . . . See 4.4.2.1

c. Hydrostatic pressure test . . . . . See 4.4.3.

4.3.2 Sampling plan and tests. Filter elements shall be subjected to the sampling plan and tests as specified in MIL-F-52308.

4.4 Test methods

4.4.1 Preproduction tests. The filter-separator shall be subjected to all the tests described herein and in MIL-F-8901, except as specified in 4.4.1.1.

4.4.1.1 Life and inhibited fuel test. If elements used in the filter-separator have been qualified under MIL-F-52308, the life and the inhibited fuel test may be waived.

4.4.2 Examination of product. The filter-separator shall be subjected to the examination of product test described in MIL-F-8901.

4.4.2.1 Canister inspection. Each separator canister shall be subjected to the permanent canister inspection test described in MIL-F-8901.

4.4.3 Hydrostatic pressure test. The filter-separator shall be subjected to a hydrostatic pressure of 275 psi in accordance with the procedures specified in MIL-F-8901. Each compartment shall be individually tested to 75 psi.

4.4.4 Element test. The filter-separator elements shall be subjected to the filter-coalescer element test specified in MIL-F-8901.

4.4.5 Flow rate. During the full-scale test conditions, the flow rate for the filter elements shall be 1,200 gpm.

4.4.6 Disassembly inspection. Upon completion of the life test specified in MIL-F-8901, the filter-separator housing shall be disassembled and inspected to determine the condition of the filter elements. Defects in the filter elements such as swelling of the element, damaged elements, damaged gaskets, et cetera, shall be noted. Excessive swelling of or damage to the element or other parts shall be cause for rejection.

4.5 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Levels A and B. One each filter-separator (with associated parts) shall be preserved and packaged in accordance with method III of MIL-P-116. All openings shall be plugged or sealed with approved materials to prevent the entry of moisture or other foreign matter.

5.1.2 Level C. Preservation and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2). Insofar as practical, shipping containers shall be uniform in size and shape; of minimum tare, weight, and cube; and shall contain only one filter-separator and associated parts.

5.2.1 Levels A and B. Unless otherwise specified (see 6.2), the shipping container shall be a type I, class 2, style B wooden crate conforming to MIL-C-104.

5.2.2 Level C. Filter-separators which require overpacking for acceptance by the carrier shall be packed in an exterior-type shipping container in a manner that will insure safe delivery at the lowest transportation rate to cited destinations. Containers shall meet Consolidated Freight rules and regulations, and rules of other common carriers as applicable to the mode of transportation.

5.3 Shipment marking. In addition to any special markings required by the contract or order (see 6.2), all interior packs and shipping containers shall be marked in accordance with MIL-STD-129. The shipment marking shall be as follows:

FILTER-SEPARATOR, LIQUID FUEL FFU-25/E

6. NOTES

6.1 Intended use. The FFU-25/E filter-separator is intended for use in airfield refueling systems, motor fuel servicing equipment, and military pipeline systems for the removal of undissolved water and solid contaminants from aviation and motor fuels.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When swing-type bolts are not to be used on all main closures of the housing (see 3.7.3.1)

c. When test samples are to consist of other than a complete filter-separator, 183 additional elements, and one new canister (see 4.2.1)

d. Special shipment marking when required (see section 5)

e. Required levels of preservation and packaging, and packing (see section 5)

Level B preservation and packaging (indicated in 5.1.1) is intended to provide economical but limited protection, and should be specified only when it is determined the filter-separators will be held in covered storage approximately 12 months from date of initial packaging.

6.3 Contract administration

6.3.1 The contracting organization should require that a prospective contractor shall have or can contract for a suitable test system prior to the award of any contracts for filtration equipment. A test system is the piping, test equipment apparatus, arrangement of test components, laboratory equipment, laboratory supplies, and techniques used during the evaluation of fuel filtration equipment.

6.3.2 The contracting organization should provide for Government inspection during the preproduction buildup and test in addition to providing Government inspection after production is released.

6.3.3 The contracting organization should, prior to issuance of a procurement bid or proposal (IFB/RFP), contact the engineering activity having technical responsibility for the specification to confirm that the latest edition of the specification is being cited.

6.4 Abbreviations. In order to save space and to avoid the spelling out of repetitious words and phrases, the following abbreviations have been used in the text of this specification:

ANSI	American National Standards Institute, Inc
ASME	American Society of Mechanical Engineers
gpm	gallons per minute
NPT	National taper pipe (thread)
psi	pounds per square inch

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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bearing housing. Seals furnished shall be suitable for use at the temperatures

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